

Playing osu! Like a Human: Cursor Generation with a TCN-WGAN and Key Press Decoding

pekin

<https://github.com/pekingfather15/osuai>

October 2026

Abstract

We train neural networks to generate human-like play for the rhythm game osu! from recordings of human players and play the generated data back in osu!lazer. The task is framed as imitation learning: the models learn the conditional distribution of a player’s cursor path and key presses given a map. Building on the open-source project by niooii, we collect 1,992 replays and compare four established deep learning sequence models (LSTM, VAE, WGAN-GP and a temporal convolutional WGAN) under one evaluation protocol. The largest gains came from correcting training objectives rather than from new architectures. We show that a finite-difference gradient penalty measures only about $1/\sqrt{d} = 1/64$ of the critic’s gradient, that a squared-error loss provably drives the cursor to the spinner center, and that the evaluation used half the real circle radius. With generator pretraining, a polar spinner head, a spin statistics loss and a capped adversarial weight, the TCN-WGAN reaches 99.89% Relax accuracy on 46 unseen maps (humans: 99.97%) and spins at 396 RPM (humans: 371). For key presses, we find that near-simultaneous double presses cause most of the accuracy lost in game; merging presses closer than 36 ms, chosen on a validation set, raises full accuracy from 93.55% to 95.45% (humans: 96.04%). Playing logged out, the system scored 96.63% on a 7.11-star map with Hidden and Hard Rock, with a hit-error UR of 81.

1 Introduction

osu! is a rhythm game in which the player moves a cursor onto circles, sliders and spinners and presses a key in time with the music. It is also where the AI streamer Neuro-sama began: before it became a chatbot VTuber, it was a program that played osu! (Neuro-sama, n.d.). That history is how this project began.

Writing a program that plays osu! perfectly is easy, since the map file specifies every object’s position and time. The harder and more interesting problem, shared with the project by niooii (2026) on which this work builds, is to make a program play *like a human*, with smooth but imperfect cursor paths, real spinning and natural key timing. This is a machine learning problem: there is no formula for “human-like”, so the behaviour has to be learned from examples of human play.

The work makes the following contributions:

- a formulation of the task as conditional generative modelling for imitation learning, and an explanation, from this formulation, of why deterministic regressors such as LSTMs cannot spin;
- a data pipeline and fixed test/validation splits over 1,992 replays, with held-out maps identified by the hash of the map file;
- an evaluation that scores generated and human plays identically, validated against in-game results;

- a diagnosis of why the original GAN models did not aim or spin, with fixes that make the TCN-WGAN the best model;
- an analysis of key press errors in game and a decoding step that removes most of them;
- playback in osu!lazer, the current open-source client (ppy, n.d.-a), restricted to logged-out play.

2 Background

2.1 The game

A map is a sequence of hit objects, each with a position on a 512×384 playfield and a time. Circles must be clicked on time, sliders must be clicked and then followed while a key is held, and spinners must be spun around the centre. The circle radius is $r = 54.4 - 4.48 \text{ CS}$ osu!pixels, where CS is the map’s circle size (ppy, n.d.-b). The hit windows depend on the overall difficulty (OD): a press within $80 - 6 \text{ OD ms}$ of the object’s time scores 300, within $140 - 8 \text{ OD ms}$ 100, and within $200 - 10 \text{ OD ms}$ 50 (ppy, n.d.-c). At OD 10 a player must press within $\pm 20 \text{ ms}$ for a 300. Accuracy is

$$\text{acc} = \frac{300 n_{300} + 100 n_{100} + 50 n_{50}}{300 (n_{300} + n_{100} + n_{50} + n_{\text{miss}})}. \quad (1)$$

Players describe timing consistency with the unstable rate (UR), ten times the standard deviation of the hit errors in milliseconds. The Relax mod presses keys automatically, so an accuracy measured with Relax reflects only the cursor.

2.2 Human movement

Two classic results describe what human cursor movement looks like. Fitts’ law states that the time to reach a target grows with $\log_2(2D/W)$, where D is the distance and W the target width (Fitts, 1954), so humans slow down for far or small targets. The minimum-jerk model shows that point-to-point arm movements follow the smoothest path, minimising the integral of the squared jerk (the derivative of acceleration), which yields smooth, bell-shaped velocity profiles (Flash & Hogan, 1985). Human movement is also variable: no two players, and no two attempts by one player, produce the same path. A human-like model must therefore produce smooth paths and a *distribution* of paths rather than a single answer.

2.3 Learning from demonstrations

Learning a behaviour from recordings of an expert is called imitation learning; its simplest form, behavioural cloning, treats it as supervised learning from observations to actions (Pomerleau, 1989; Hussein et al., 2017). Here the observation is the map x (a sequence of features per time step) and the action is the player’s cursor path and key state y . A known weakness of behavioural cloning is covariate shift: when the agent’s own mistakes lead it to states the expert never visited, errors compound (Ross et al., 2011). Our setting avoids this, because generation is open-loop: the whole play is generated from the map before the game starts, and the game state does not feed back into the model.

What a model can learn depends on its loss. A model trained with the squared error learns the conditional mean:

$$\arg \min_f \mathbb{E} [\|y - f(x)\|^2] = \mathbb{E}[y \mid x] \quad (2)$$

(Bishop, 2006, Section 1.5.5). When $p(y \mid x)$ has a single mode, the mean is a good answer, which is why regressors aim well. When it has many modes, the mean can be an answer no human would give. On a spinner, players draw circles of different phases around the centre; averaging over the phase ϕ , $\mathbb{E}_\phi[c + r(\cos \phi, \sin \phi)] = c$, the centre itself. A deterministic regressor therefore sits still on

spinners. To be human-like a model has to *sample* from $p(y \mid x)$, which is what generative models do.

3 Models

All models are deep neural networks trained by gradient descent. Section 3.5 compares them with current models.

3.1 Recurrent networks and LSTM

A recurrent network reads a sequence one step at a time and keeps a hidden state h_t that summarises the past. Plain recurrent networks are hard to train over long sequences because gradients shrink or explode as they are propagated back through many steps. The long short-term memory (LSTM) adds a cell state c_t that is updated additively and controlled by gates (Hochreiter & Schmidhuber, 1997; Gers et al., 2000):

$$\begin{aligned} i_t, f_t, o_t &= \sigma(W[h_{t-1}, x_t] + b), & \tilde{c}_t &= \tanh(W_c[h_{t-1}, x_t] + b_c), \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, & h_t &= o_t \odot \tanh(c_t), \end{aligned} \quad (3)$$

where the input, forget and output gates i_t, f_t, o_t decide what is written, kept and read. LSTMs have generated human handwriting from pen trajectories (Graves, 2013), a task close to ours. Trained with a squared error (or the Huber loss used here, which is quadratic for small errors), however, an LSTM is a regressor and outputs the mean path of Eq. (2).

3.2 Temporal convolutional networks

A temporal convolutional network (TCN) replaces recurrence with stacked one-dimensional convolutions. A causal convolution only uses inputs up to time t ; a dilated convolution with kernel size k and dilation d looks at inputs $t, t-d, \dots, t-(k-1)d$ (Yu & Koltun, 2016; van den Oord et al., 2016). Doubling the dilation in each layer grows the receptive field exponentially with depth:

$$R = 1 + (k-1) \sum_{\ell=1}^L d_\ell, \quad d_\ell = 2^{\ell-1}. \quad (5)$$

TCNs process all time steps in parallel and have been found to match or outperform LSTMs on many sequence tasks while keeping a longer effective memory (Bai et al., 2018). Because the whole map is known in advance, we can also shift some layers to look into the future. Convolutions also support the second-order gradients that the gradient penalty needs, which an LSTM in cuDNN does not (Section 5.1).

3.3 Variational autoencoders

A variational autoencoder (VAE) learns a latent variable model $p_\theta(y \mid z, x)$ with a prior $p(z)$. Since the likelihood cannot be computed directly, an encoder $q_\phi(z \mid y, x)$ approximates the posterior, and both are trained to maximise the evidence lower bound (Kingma & Welling, 2014; Rezende et al., 2014):

$$\log p_\theta(y \mid x) \geq \mathbb{E}_{q_\phi(z \mid y, x)} [\log p_\theta(y \mid z, x)] - \text{KL}(q_\phi(z \mid y, x) \parallel p(z)). \quad (6)$$

Sampling z gives different outputs for the same map. With a powerful decoder, however, the model can ignore z and drive the KL term to zero; this *posterior collapse* (Bowman et al., 2016) turns the VAE back into a mean predictor. Adversarial VAEs add a discriminator to improve realism (Plummer et al., 2020).

3.4 Generative adversarial networks and WGAN-GP

A generative adversarial network (GAN) trains a generator $G(z, x)$ against a discriminator that tries to tell real data from generated data (Goodfellow et al., 2014); conditioning both on x gives a conditional GAN (Mirza & Osindero, 2014). The original objective is unstable and prone to mode collapse, in which the generator covers only a few modes. The Wasserstein GAN instead minimises the earth mover’s distance, which by the Kantorovich–Rubinstein duality is

$$W(p_r, p_g) = \sup_{\|D\|_L \leq 1} \mathbb{E}_{y \sim p_r}[D(y)] - \mathbb{E}_{y \sim p_g}[D(y)], \quad (7)$$

where the critic D must be 1-Lipschitz (Arjovsky et al., 2017). WGAN-GP enforces this with a gradient penalty on random interpolates $\hat{y}$ between real and generated samples (Gulrajani et al., 2017):

$$\mathcal{L}_{\text{GP}} = \lambda \mathbb{E}_{\hat{y}} \left[(\|\nabla_{\hat{y}} D(\hat{y})\|_2 - 1)^2 \right]. \quad (8)$$

Computing $\nabla_{\theta} \mathcal{L}_{\text{GP}}$ requires differentiating a gradient (double backpropagation). Adversarial losses are often combined with a reconstruction loss that keeps outputs close to the target, as in image-to-image translation (Isola et al., 2017); we do the same.

3.5 Relation to the state of the art

None of these architectures is new: LSTM dates from 1997, VAE and GAN from 2014, WGAN-GP from 2017 and the TCN as evaluated here from 2018. They are established methods, chosen because they fit a dataset of 10,730 training segments and one laptop GPU, and because the original project implemented them. The state of the art in sequence and motion generation has since moved to Transformers (Vaswani et al., 2017), denoising diffusion models (Ho et al., 2020), which generate human body motion with high realism (Tevet et al., 2023), and state-space models such as Mamba (Gu & Dao, 2023). These models are larger and more data hungry. This work does not propose a new architecture. It analyses why the established ones failed on this task and fixes them using the principles above.

3.6 Related work on osu!

OsuLearn trained a recurrent network on replays to generate cursor movement (GuiBrandt, n.d.). niooii (2026) extended it with a VAE, a WGAN-GP with an LSTM critic and a TCN-WGAN, reporting Relax accuracies of 98.10% (LSTM), 98.89% (VAE), 99.68% (LSTM-WGAN) and 99.40% (TCN-WGAN) on one map in a video. The author notes that the LSTM is “too good” and robotic, that the VAE averages movement and cannot reliably spin, and that the TCN variant “is probably not set up well”. Keys come from a separate LSTM that “sometimes will tap twice for trickier sequences”; this paper finds that this double tapping is the main source of lost accuracy in game. Other open-source projects generate cursor movement for an editor (minetoblend, n.d.) or play from screen captures with object detection (gameswu, n.d.).

4 Data

Sources. Replays of the players ranked first and third in the world were downloaded from the official website (671 replays). To cover more players we imported the osu!3k dataset (ilymeow, n.d.) (MIT licence), selecting replays by player performance points with accuracy $\geq 90\%$ and star rating ≥ 5 , and excluding mods that change the game (Relax, Autopilot, Spun Out, Easy, Half Time). Maps were downloaded from a mirror and matched to replays by the MD5 hash of the map file.

Table 1: Dataset versions. A segment is 2,048 frames of 12 ms.

Version	Training replays	Segments
V1	212 (two players)	1,409
V2	425 (two players)	2,530
V3	1,992 (official 671 + osu!3k 1,410; over 1,000 players)	10,730
niooii (2026)	–	13,617

Cleaning. Some osu!3k entries are the same play rendered twice with different mods; all such conflicting duplicates were dropped. 20 replays by players with under 2,000 total pp but high-star, high-accuracy scores were excluded as implausible.

Splits. The test set holds 50 replays (46 usable) downloaded after V1 was trained, on maps never used for training, one replay per map. A validation set of 30 replays (27 usable) is used only to choose checkpoints and decoding settings. From V3 on, test and validation maps are excluded by file hash rather than path, which caught 9 more replays of the same maps stored in other folders.

Representation. Each map becomes a sequence of 12 ms frames with 9 features: the position of the next object, the time until it, the object type, CS, and slider speed and length. The target is the cursor position and the state of the two keys. Sequences are cut into chunks of 2,048 frames (24.6 s).

5 Method

5.1 TCN-WGAN cursor model

The generator is a TCN with kernel size 3 over the map features concatenated with an 8-dimensional noise sequence. Its first six layers (dilations 1 to 32) each shift the kernel two frames into the future, so the generator sees 12 frames (144 ms) ahead, and its last four layers are causal; in total it sees 1.7 s of the past. The critic is a TCN over the map features and a cursor trajectory whose seven dilated layers each look one frame ahead, and outputs one score per sequence. The generator has 277,956 parameters, the critic 141,025. The generator loss is

$$\mathcal{L}_G = \mathcal{L}_{\text{pos}} - \lambda_{\text{adv}} \mathbb{E}[D(G(z, x), x)] + \lambda_{\text{spin}} \mathcal{L}_{\text{spin}}, \quad (9)$$

where $\mathcal{L}_{\text{pos}}$ is the Huber (smooth L1) position error, which is quadratic for errors below 1 and therefore behaves like the squared error of Eq. (2) when aiming. The critic uses the exact penalty of Eq. (8) with $\lambda = 8$ and three critic steps per generator step. We use Adam (Kingma & Ba, 2015) with $\beta = (0.5, 0.9)$.

Why the original penalty did not work. The original WGAN code replaced Eq. (8) with a finite difference along one random unit direction u , $(D(\hat{y} + \epsilon u) - D(\hat{y}))/\epsilon \approx u^\top g$ with $g = \nabla D(\hat{y})$, to avoid double backpropagation through the LSTM critic. For u uniform on the unit sphere in $\mathbb{R}^d$, $\mathbb{E}[(u^\top g)^2] = \|g\|^2/d$, so $|u^\top g| \approx \|g\|/\sqrt{d}$. A trajectory of 2,048 frames with (x, y) has $d = 4,096$ dimensions, so the penalty sees only 1/64 of the gradient and pushes $\|g\|$ towards 64 instead of 1. The critic is then far from 1-Lipschitz; in our runs its loss stalled near 3.66 and the Wasserstein estimate stayed near 0, meaning it could not separate real from generated play. The exact penalty with an LSTM critic took 12.2 s per step (cuDNN’s LSTM does not support double backward) against 0.13 s for the approximation, which is impractical; a convolutional critic computes it cheaply, which is why we use the TCN.

Pretraining. The original learning rate of 10^{-4} was still reducing the position error after 100 epochs. We first train the generator for 60 epochs on $\mathcal{L}_{\text{pos}}$ alone at 10^{-3} , then train adversarially at 10^{-4} , ramping λ_{adv} up to at most 2×10^{-4} . A larger weight let the critic’s scores grow and pulled the cursor away from the targets.

Spinners. Following Eq. (2), a position loss on spinner frames pulls the cursor to the centre, so we remove spinner frames from $\mathcal{L}_{\text{pos}}$. A separate polar head outputs a radius r_t and an angular speed ω_t during spinners:

$$(x_t, y_t) = c + r_t (\cos \theta_t, \sin \theta_t), \quad \theta_t = \theta_0 + s \sum_{\tau \leq t} \omega_\tau, \quad (10)$$

where c is the playfield centre and the start angle θ_0 and direction $s \in \{-1, 1\}$ come from the noise. Separating the head keeps spinner training from interfering with aiming. The spin statistics loss asks for at least the players’ mean angular speed $\bar{\omega}$ and a similar mean radius $\bar{r}$, but not for a particular phase, so it has no averaging problem:

$$\mathcal{L}_{\text{spin}} = \max(0, \bar{\omega}_{\text{real}} - \bar{\omega}_{\text{fake}}) + \left(\frac{\bar{r}_{\text{fake}} - \bar{r}_{\text{real}}}{100} \right)^2, \quad (11)$$

with $\lambda_{\text{spin}} = 0.02$ from epoch 30.

5.2 Key model and decoding

Keys come from the original key model: a two-layer LSTM (64 units) and a dense layer that predict, for each frame, the probability that each of the two keys is held (54,626 parameters), trained with binary cross-entropy against the human key state. Two properties of this formulation explain its errors. First, a press lasts one frame while a hold lasts five to ten, so the loss is dominated by hold frames and barely changes when the press moment is off by a frame; press events are rare, a class imbalance familiar from object detection (Lin et al., 2017). Second, the model is often unsure which key a human would use, so both probabilities hover near 0.5 and thresholding each key separately presses both within a frame or two (Fig. 1). In game every extra press can hit the next object early, and in osu!lazer only the key that hit a slider head keeps tracking the slider, so a short extra press that reaches the head first and lets go loses the slider tail.

We therefore decode presses: presses that start less than g apart are merged into one, held until the last of them is released, and each press gets a key that is not already held, releasing a key for one frame before pressing it again. Merging near-duplicate detections is the idea behind non-maximum suppression in object detection (Neubeck & Van Gool, 2006). The gap g is chosen on the validation set.

5.3 Playback in osu!lazer

The original playback reads osu!stable’s memory. For osu!lazer we read the game state from the websocket of tosu (tosu, n.d.): the map file, the song time, the mods and the profile. The song time arrives every 10–20 ms, so it is extrapolated with the local clock between updates and never allowed to step back by less than 50 ms. The cursor is mapped to the screen with lazer’s playfield transform (scale $h/480$, centred). Key changes of skipped frames are replayed in order, and an offset compensates for the playback latency. A small window generates the play when a map is selected, can generate the cursor alone for testing with the Relax mod, and enables playback only when tosu reports that nobody is logged in.

5.4 Implementation

The V3 models were trained with the project’s first implementation, which was built on the code of niooi (2026). All code in the repository was then rewritten for this project: map and replay

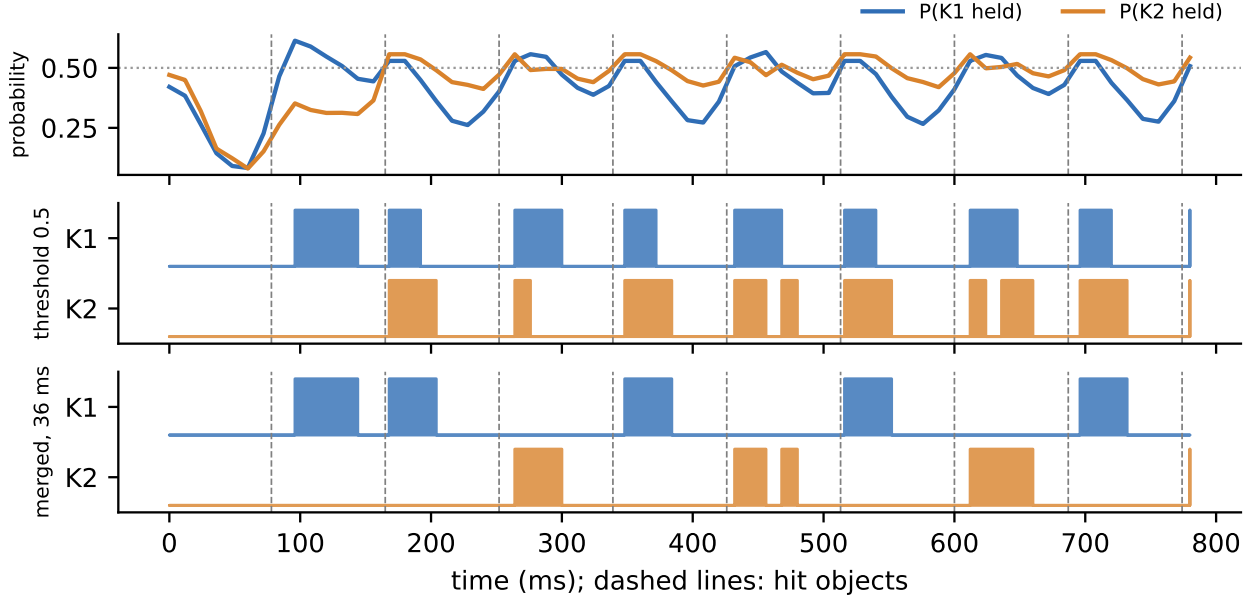


Figure 1: Key model output in an 800 ms stream on a 7.88-star map. Top: hold probabilities. Middle: thresholding each key at 0.5 presses both keys at almost every object. Bottom: after merging presses closer than 36 ms. Presses also come slightly after the objects (7 ms on average over the map), a delay the playback offset compensates.

parsing (including the slider curves, which now follow the curve types of the .osu format instead of a coarse approximation), the features, the models, training, evaluation and playback. The V3 weights were converted to the new code and give identical outputs for identical inputs. The results of Sections 7.1 and 7.2 come from the first implementation; the key decoding results and Table 4 were recomputed with the new code.

6 Evaluation

For every test replay the model generates a play for the same map and mods, and the generated and human plays are scored the same way:

- **Relax accuracy:** each circle or slider head is judged by the closest moment within its 50 window at which the cursor is inside the circle, as if the game pressed the keys.
- **Full accuracy:** judged with the key presses. Each press of either key can hit one object; an object takes the earliest unused press in its window while the cursor is inside.
- **Slider tails:** the key that hit the head must stay held, and the cursor inside the follow circle ($2.4r$), until 36 ms before the slider ends.
- **Spinner RPM:** rotations around the playfield centre during spinners.

The original evaluation code used half the real circle radius; all numbers here use the corrected radius unless stated. Full accuracy underestimates real accuracy: the human replays score 95.84% here but 99.00% according to the replay files, mostly because 12 ms sampling loses some short presses. The evaluation is nevertheless predictive: for the 7.88-star map below it predicts 90.63% and 391/420 slider tails, and the game reported 90.09% and 393/420. The rewritten evaluation reproduces the first one: on the same 46 test plays the V3 TCN-WGAN scores 93.44% full accuracy without and 95.47% with press merging (first implementation: 93.41% and 95.49%), with identical Relax accuracy (99.89%), hit rate, mean distance and spinner speed. It can also read 4 test maps the first parser could not, so the tables below use all 50.

Table 2: Test set (46 unseen maps), first implementation, corrected radius, all with the V3 key model (no decoding, first version of the full-accuracy judge).

Source	Full acc.	Relax acc.	Relax misses	Mean dist.	Spinner
Human	95.84%	99.97%	5	11.9 px	371 RPM
V1 LSTM	94.94%	99.95%	14	9.1 px	3
V1 WGAN-GP	62.53%	80.49%	4,988	33.7 px	41
V2 LSTM	94.79%	99.96%	13	9.7 px	6
V2 VAE	51.92%	71.13%	8,219	39.1 px	42
V2 WGAN-GP	81.55%	91.95%	2,482	24.3 px	34
V2 TCN-WGAN (original settings)	73.19%	91.17%	1,962	26.3 px	29
V3 LSTM	94.76%	99.92%	27	10.6 px	3
V3 TCN-WGAN	94.68%	99.89%	28	8.3 px	396

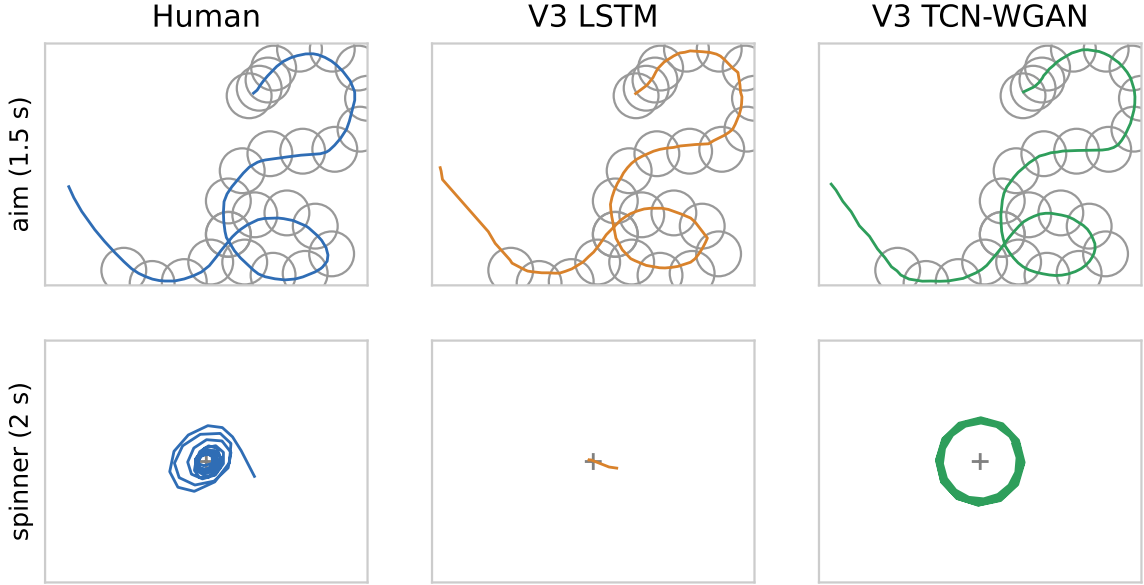


Figure 2: Cursor paths on an unseen map. Top: 1.5 s of aiming. Bottom: the first 2 s of a spinner. The LSTM aims as well as the GAN but stays at the spinner centre, as Eq. (2) predicts.

7 Results

7.1 Cursor models

Table 2 compares all models on the 46 test maps with the same key model, and Fig. 2 shows their trajectories.

The results follow the principles of Section 2. The LSTMs aim almost perfectly, as expected from a regressor on a single-mode target, but never spin (3–6 RPM). The VAE’s KL term fell to near zero in every version, a posterior collapse that leaves it a mean predictor, and its spinner loss is again a position loss. The original GAN models aim poorly, which we trace to the penalty and the learning rate. After the fixes the V3 TCN-WGAN aims as precisely as the LSTM and spins like a human. More diverse data (V3, over a thousand players) made the LSTM slightly worse, as its mean path becomes an average of more styles, but helped the GAN, which samples from the distribution of styles.

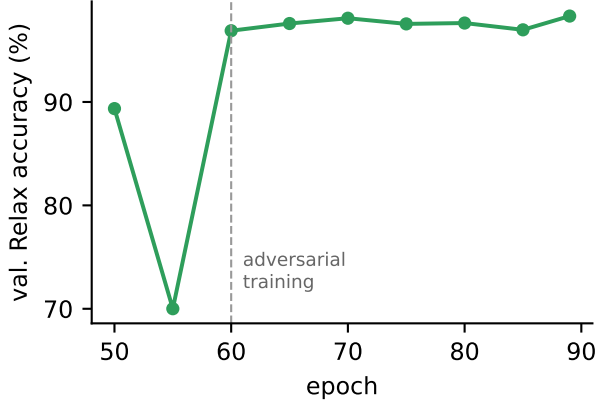
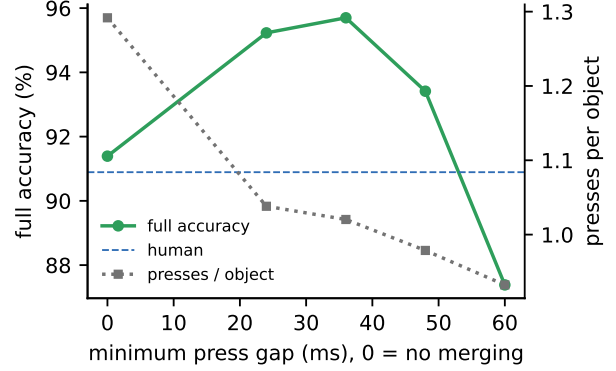
7.2 Ablation

Table 3 adds the fixes one at a time on V2 data. These numbers use the old, halved radius and are only comparable with each other.

Pretraining alone raised Relax accuracy from 56.5% to 96.7%. Sharing the output between

Table 3: Ablation of the TCN-WGAN on V2 data (test set, old halved radius).

Change	Relax acc.	Mean dist.	Spinner
Original settings (100 epochs, lr 10^{-4})	56.5%	26.6 px	24 RPM
+ generator pretraining (40 epochs, lr 10^{-3})	96.7%	10.1 px	26
+ no position loss on spinners, spin loss (shared output)	89.9%	14.7 px	86
+ polar spinner head	89.3%	14.5 px	372
+ 60 pretraining epochs, $\lambda_{\text{adv}} \leq 2 \times 10^{-4}$	97.4%	9.5 px	519

**Figure 3:** V3 TCN-WGAN checkpoints on the validation set (old radius). Epoch 89 was chosen.**Figure 4:** Choosing the press merging gap on the validation set.

aiming and spinning cost accuracy, which the separate polar head and the lower adversarial weight recovered. Accuracy varies a lot between checkpoints (70% at epoch 55, 97% at epoch 60; Fig. 3), as is common in adversarial training, so checkpoints must be chosen on a validation set.

7.3 Key decoding

The judge was updated to count each key’s presses separately and to score slider tails. Fig. 4 and Table 4 show that merging presses closer than 36 ms brings the presses per object from 1.29 to 1.02 (humans: 1.00) and full accuracy above the human replays on the validation set. Larger gaps start merging real presses in fast streams.

Merging removes most 50s but adds misses (323 to 770 on the test set): some real presses are merged away. The V3 LSTM, scored the same way, reaches 95.74% with merging; its keys come from the same key model, and it aims slightly more precisely, but it does not spin (3 RPM). Decoding cannot fully fix a model that was trained to predict held states rather than press events.

8 In-Game Results

All runs were played in osullazer while logged out (guest), full screen at 1920×1080 , with the V3 TCN-WGAN and the V3 key model (Table 5). Hit errors were recorded from tosu (Fig. 5).

The first run on the 7.88-star map (OD 9.6, ± 22 ms for a 300) lost most 300s to a systematic delay of about 14 ms. Offline, adding a 7 ms delay and 15 ms jitter to the presses reproduced the game result almost exactly (73.80%, 646/328/42/17), which shows how sensitive a narrow window is to timing noise. Calibrating the offset raised it to 90.09%. The remaining 50s and the 27 lost slider tails came from double presses: simulating the game’s rules offline lost exactly 27 tails. After merging, the 7.11-star map with Hard Rock (OD 10, ± 20 ms) scored 96.63% without a single 50, with a hit-error standard deviation of 8.1 ms (UR 81), against a UR of 209 and 243 in the two earlier recorded runs.

Table 4: Key decoding, V3 TCN-WGAN with the V3 key model, scored with the rewritten evaluation (30 validation and 50 test plays).

Set	Decoding	Full acc.	300/100/50/miss	Slider tails	Presses/obj.
Validation	human	90.89%	20906/1482/31/1131	94.3%	1.00
	threshold 0.5	91.39%	20856/1555/890/249	94.5%	1.29
	merged, 24 ms	95.23%	22097/881/215/357	96.4%	1.04
	merged, 36 ms	95.70%	22270/739/124/417	96.4%	1.02
Test	human	96.04%	37150/2244/31/40	99.4%	1.03
	threshold 0.5	93.55%	36052/2108/982/323	95.3%	1.16
	merged, 36 ms	95.45%	37259/1031/405/770	95.8%	1.00

Table 5: In-game results in osu!lazer (guest). Maps by Ado (first three) and Kenshi Yonezu. Offset: playback latency compensation in ms; merged: presses closer than 36 ms merged. Offline: prediction of the evaluation for the same map.

Map	Stars	Settings	Accuracy	300/100/50/miss	Offline
AiAiA [Kiss me Insane]	4.07	offset 15	98.38%	668/12/2/3	–
AiAiA [Be loved Expert]	5.70	offset 15	95.85%	820/39/15/3	97.30% [†]
Aishite Aishite Aishite [</3]	7.88	offset 15	74.88%	579/394/40/21	–
same map	7.88	offset 29	90.09%	876/69/68/21	90.63%
IRIS OUT [Darling!!!]	7.11	HD HR, 29, merged	96.63%	654/28/0/5	97.96%

[†] first version of the judge.

9 Discussion

Lessons. (1) An approximation of a training objective can silently break the method: the finite-difference penalty measured $1/\sqrt{d}$ of the gradient, and the critic never learned. (2) An optimiser setting can matter more than the architecture: pretraining at a higher learning rate turned 56.5% into 96.7%. (3) The loss decides what the model can represent: a squared error cannot produce a multi-modal behaviour such as spinning, and the VAE’s spinner loss had the same flaw. (4) Too much adversarial weight trades accuracy for realism. (5) For keys, predicting held states per frame makes the press moment unimportant to the loss and invites double presses. (6) The evaluation must match the game: a halved radius and a judge that merged the two keys hid real problems, and once corrected, offline predictions matched the game within about one percentage point.

Limitations. The full-accuracy judge underestimates humans and does not model note lock or slider ticks. Human-likeness is argued from trajectories and statistics (spinner speed, presses per object, UR), not measured with a perceptual study. The playback latency drifts by about 10 ms between sessions. Merging adds misses. Only five in-game runs were recorded. V3 still has 79% of the original author’s data.

Future work. A key model that predicts press events with a sub-frame offset, with a focal or weighted loss for the rare events (Lin et al., 2017), a release tied to the end of the object it hit, a non-causal TCN, and the generated cursor as input, should remove double presses without merging away real ones. A single model for cursor and keys would let presses depend on where the cursor is. Diffusion models (Ho et al., 2020; Tevet et al., 2023) are the natural next step for the cursor model if more data becomes available.

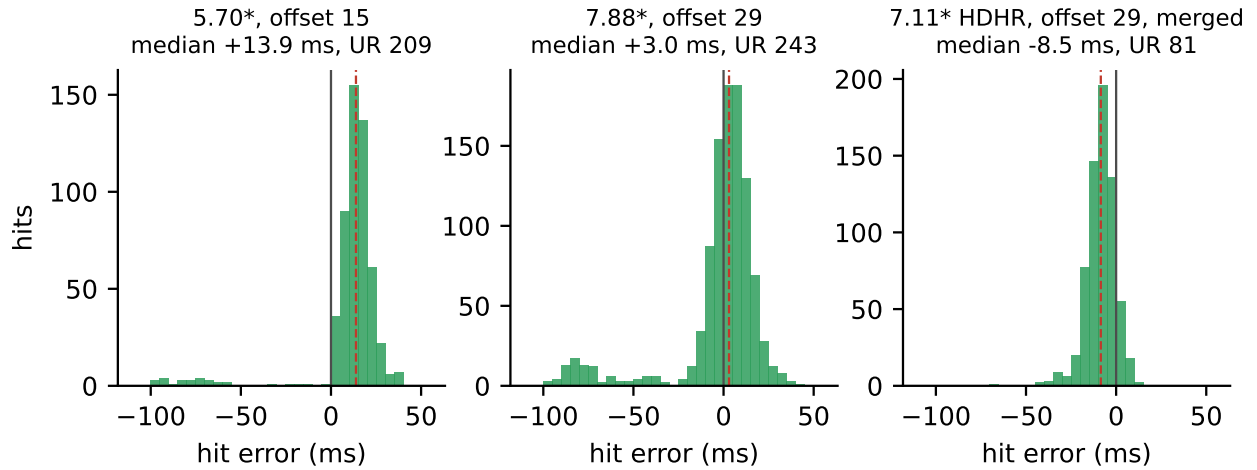


Figure 5: Hit errors recorded in game (positive = late). Left: presses about 14 ms late before calibrating the offset. Middle: centred, but with a tail of presses 60–100 ms early (one beat early at 230 BPM). Right: after merging the tail is gone and UR is 81.

10 Ethics

All in-game experiments were played while logged out of *osu!*, so nothing was submitted to the online leaderboards; using such playback on an online account breaks the game’s rules and is unfair to other players. The playback has no anti-detection features. Music and map files are not redistributed. The *osu!3k* dataset is used under its MIT licence.

Acknowledgements

Thanks to niooi for the original project and models, to GuiBrandt for *OsuLearn*, to the authors of *osu!3k*, and to the developers of *tosu* and *osu!lazer*. The code, experiments and this paper were developed with the help of Claude (Anthropic) as a coding assistant. The code is available at <https://github.com/pekingfather15/osuai>.

References

- Arjovsky, M., Chintala, S., & Bottou, L. (2017). Wasserstein generative adversarial networks. In D. Precup & Y. W. Teh (Eds.), *Proceedings of the 34th International Conference on Machine Learning* (Vol. 70, pp. 214–223). PMLR. <https://proceedings.mlr.press/v70/arjovsky17a.html>
- Bai, S., Kolter, J. Z., & Koltun, V. (2018). *An empirical evaluation of generic convolutional and recurrent networks for sequence modeling* [Preprint]. arXiv. <https://doi.org/10.48550/arXiv.1803.01271>
- Bishop, C. M. (2006). *Pattern recognition and machine learning*. Springer.
- Bowman, S. R., Vilnis, L., Vinyals, O., Dai, A. M., Jozefowicz, R., & Bengio, S. (2016). Generating sentences from a continuous space. In *Proceedings of the 20th SIGNLL Conference on Computational Natural Language Learning* (pp. 10–21). Association for Computational Linguistics. <https://doi.org/10.18653/v1/K16-1002>
- Fitts, P. M. (1954). The information capacity of the human motor system in controlling the amplitude of movement. *Journal of Experimental Psychology*, 47(6), 381–391. <https://doi.org/10.1037/h0055392>
- Flash, T., & Hogan, N. (1985). The coordination of arm movements: An experimentally confirmed mathematical model. *The Journal of Neuroscience*, 5(7), 1688–1703. <https://doi.org/10.1523/JNEUROSCI.05-07-01688.1985>
- gameswu. (n.d.). *AutOsu* [Computer software]. GitHub. Retrieved October 7, 2026, from <https://github.com/gameswu/AutOsu>
- Gers, F. A., Schmidhuber, J., & Cummins, F. (2000). Learning to forget: Continual prediction with LSTM. *Neural Computation*, 12(10), 2451–2471. <https://doi.org/10.1162/089976600300015015>
- Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., & Bengio, Y. (2014). Generative adversarial nets. In *Advances in Neural Information Processing Systems* (Vol. 27). Curran Associates.
- Graves, A. (2013). *Generating sequences with recurrent neural networks* [Preprint]. arXiv. <https://doi.org/10.48550/arXiv.1308.0850>
- Gu, A., & Dao, T. (2023). *Mamba: Linear-time sequence modeling with selective state spaces* [Preprint]. arXiv. <https://doi.org/10.48550/arXiv.2312.00752>
- GuiBrandt. (n.d.). *OsuLearn* [Computer software]. GitHub. Retrieved October 7, 2026, from <https://github.com/GuiBrandt/OsuLearn>
- Gulrajani, I., Ahmed, F., Arjovsky, M., Dumoulin, V., & Courville, A. C. (2017). Improved training of Wasserstein GANs. In *Advances in Neural Information Processing Systems* (Vol. 30). Curran Associates.
- Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems* (Vol. 33, pp. 6840–6851). Curran Associates.
- Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>
- Hussein, A., Gaber, M. M., Elyan, E., & Jayne, C. (2017). Imitation learning: A survey of learning methods. *ACM Computing Surveys*, 50(2), Article 21. <https://doi.org/10.1145/3054912>
- ilymeow. (n.d.). *osu!3k* [Data set]. Kaggle. Retrieved October 7, 2026, from <https://www.kaggle.com/datasets/ilymeow/osu3k>
- Isola, P., Zhu, J.-Y., Zhou, T., & Efros, A. A. (2017). Image-to-image translation with conditional adversarial networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 1125–1134). IEEE. <https://doi.org/10.1109/CVPR.2017.632>
- Kingma, D. P., & Ba, J. (2015). Adam: A method for stochastic optimization. In *3rd International Conference on Learning Representations*. <https://doi.org/10.48550/arXiv.1412.6980>
- Kingma, D. P., & Welling, M. (2014). Auto-encoding variational Bayes. In *2nd International Conference on Learning Representations*. <https://doi.org/10.48550/arXiv.1312.6114>
- Lin, T.-Y., Goyal, P., Girshick, R., He, K., & Dollár, P. (2017). Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision* (pp. 2980–2988). IEEE. <https://doi.org/10.1109/ICCV.2017.324>
- minetoblend. (n.d.). *osucad-cursor-predict* [Computer software]. GitHub. Retrieved October 7, 2026, from

- <https://github.com/minetoblend/osucad-cursor-predict>
- Mirza, M., & Osindero, S. (2014). *Conditional generative adversarial nets* [Preprint]. arXiv. <https://doi.org/10.48550/arXiv.1411.1784>
- Neubeck, A., & Van Gool, L. (2006). Efficient non-maximum suppression. In *18th International Conference on Pattern Recognition* (Vol. 3, pp. 850–855). IEEE. <https://doi.org/10.1109/ICPR.2006.479>
- Neuro-sama. (n.d.). In *Wikipedia*. Retrieved October 7, 2026, from <https://en.wikipedia.org/wiki/Neuro-sama>
- niooii. (2026). *osu* [Computer software]. GitHub. <https://github.com/niooii/osu>
- Plumerault, A., Le Borgne, H., & Hudelot, C. (2020). *AVAE: Adversarial variational auto encoder* [Preprint]. arXiv. <https://doi.org/10.48550/arXiv.2012.11551>
- Pomerleau, D. A. (1989). ALVINN: An autonomous land vehicle in a neural network. In D. S. Touretzky (Ed.), *Advances in Neural Information Processing Systems* (Vol. 1, pp. 305–313). Morgan Kaufmann.
- ppy. (n.d.-a). *osu!* [Computer software]. GitHub. Retrieved October 7, 2026, from <https://github.com/pppy/osu>
- ppy. (n.d.-b). Circle size. In *osu! wiki*. Retrieved October 7, 2026, from https://osu.ppy.sh/wiki/Beatmap/Circle_size
- ppy. (n.d.-c). Overall difficulty. In *osu! wiki*. Retrieved October 7, 2026, from https://osu.ppy.sh/wiki/Beatmap/Overall_difficulty
- Rezende, D. J., Mohamed, S., & Wierstra, D. (2014). Stochastic backpropagation and approximate inference in deep generative models. In *Proceedings of the 31st International Conference on Machine Learning* (Vol. 32, pp. 1278–1286). PMLR.
- Ross, S., Gordon, G., & Bagnell, D. (2011). A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics* (Vol. 15, pp. 627–635). PMLR.
- Tevet, G., Raab, S., Gordon, B., Shafir, Y., Cohen-Or, D., & Bermano, A. H. (2023). Human motion diffusion model. In *The Eleventh International Conference on Learning Representations*. <https://doi.org/10.48550/arXiv.2209.14916>
- tosu. (n.d.). *tosu* [Computer software]. GitHub. Retrieved October 7, 2026, from <https://github.com/tosuapp/tosu>
- van den Oord, A., Dieleman, S., Zen, H., Simonyan, K., Vinyals, O., Graves, A., Kalchbrenner, N., Senior, A., & Kavukcuoglu, K. (2016). *WaveNet: A generative model for raw audio* [Preprint]. arXiv. <https://doi.org/10.48550/arXiv.1609.03499>
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems* (Vol. 30). Curran Associates.
- Yu, F., & Koltun, V. (2016). Multi-scale context aggregation by dilated convolutions. In *4th International Conference on Learning Representations*. <https://doi.org/10.48550/arXiv.1511.07122>